

Contestant Number: _____

Time: _____

Rank: _____



MANAGERIAL ACCOUNTING (135) State – 2020

Multiple Choice & Short Answer Section:

Multiple Choice (15 @ 2 points) _____ (30 points)

Short Answers (10 @ 1 point) _____ (10 points)

Production Portion:

Problem 1: Variable Income Statement _____ (28 points)

Problem 2: Budgets _____ (35 points)

Problem 3: Break Even Analysis _____ (15 points)

TOTAL POINTS _____ ***(118 points)***

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

No more than ten (10) minutes orientation
No more than ninety (90) minutes testing time
No more than ten (10) minutes wrap-up

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General Instructions

You have been hired as a Financial Assistant and will be keeping the accounting records for Professional Business Associates, located at 5454 Cleveland Avenue, Columbus, Ohio 43231. Professional Business Associates provides accounting and other financial services for clients. You will complete jobs for Professional Business Associates' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts. The multiple choice and short answer questions should be completed first, and then the accounting problems may be completed in any order.

Your name and/or school name should **not** appear on any work you submit for grading. Write your contestant number in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

Percents rounded to two decimal places or the tenth percent (Ex: 22.8%)

Complete the following:

- _____ 1. Which of the following costs remain constant regardless of the level of activity?
- a. Fixed costs
 - b. Mixed costs
 - c. Opportunity costs
 - d. Variable costs
- _____ 2. Which of the following costs changes in direct proportion with changes in the level of activity?
- a. Fixed costs
 - b. Mixed costs
 - c. Opportunity costs
 - d. Variable costs

Mankin Company manufactures industrial shelving. The table below shows the factory overhead mixed costs for the last six months. Use the High Low Method to calculate the following:

Month	Total Units	Total Cost
January	10,000	\$ 100,000
February	7,000	85,000
March	12,000	110,000
April	15,000	125,000
May	11,000	105,000
June	8,000	90,000

- _____ 3. Calculate the variable cost per unit (Round to the nearest cent).
- a. \$ 5.00
 - b. \$ 8.33
 - c. \$10.00
 - d. \$12.14
- _____ 4. Calculate the fixed cost (Round to the nearest cent).
- a. \$ 5,000
 - b. \$20,000
 - c. \$40,000
 - d. \$50,000

- _____ 5. The contribution margin is . . .
- a. The same as gross profit.
 - b. The excess of sales minus fixed costs.
 - c. The excess of sales minus variable costs.
 - d. The net income.
- _____ 6. Break-even in units is calculated with the following formula . . .
- a. $\frac{\text{Total Fixed Costs}}{\text{Unit Contribution Margin}}$
 - b. $\frac{\text{Total Fixed Costs}}{\text{Contribution Margin Ratio}}$
 - c. $\frac{\text{Total Fixed Costs}}{\text{Unit Selling Price}}$
 - d. $\frac{\text{Total Fixed Costs}}{\text{Unit Variable Cost}}$
- _____ 7. If fixed costs are \$182,000, the unit selling price is \$148, and the unit variable costs is \$96, what is the break-even in units (Round to the nearest whole unit)?
- a. 1,230 units
 - b. 1,282 units
 - c. 1,896 units
 - d. 3,500 units
- _____ 8. Which type of budget allows for adjustments in activity levels?
- a. Continuous budget
 - b. Flexible budget
 - c. Static budget
 - d. Zero-based budget
- _____ 9. The master budget always begins with which budget?
- a. Income Statement budget
 - b. Materials budget
 - c. Production budget
 - d. Sales budget
- _____ 10. An unfavorable variance occurs when . . .
- a. Actual costs are less than standard costs.
 - b. Actual costs are more than standard costs.
 - c. Standard costs are more than actual costs.
 - d. None of the above.

_____ 11. Direct materials variances will include . . .

- a. Price and quantity variances.
- b. Price and time variances.
- c. Rate and quantity variances.
- d. Rate and time variances.

The following data relate to direct labor costs for the April.

Standard costs	10,200 hours at \$15
Actual costs	10,500 hours at \$14.50

_____ 12. Determine the efficiency(time) variance.

- a. \$4,500 Favorable
- b. \$4,500 Unfavorable
- c. \$5,250 Favorable
- d. \$5,250 Unfavorable

_____ 13. Determine the rate variance.

- a. \$4,500 Favorable
- b. \$4,500 Unfavorable
- c. \$5,250 Favorable
- d. \$5,250 Unfavorable

_____ 14. Which of the following explains sales mix?

- a. Sales mix determines how a company can sell its products.
- b. Sales mix is a theory of expanding your selling territories.
- c. Sales mix is the percentage distribution of products sold.
- d. Sales mix compares the sales between your business and the industry average.

_____ 15. Who would require the preparation of the Variable Income Statement?

- a. Management
- b. GAAP (Generally Accepted Accounting Principles)
- c. SEC (Securities and Exchange Commission)
- d. All of the above

The following is a list of various costs of producing couches. Classify each cost as (V) Variable, (F) Fixed, or (M) Mixed.

_____ 16. Wood used to build the frame.

_____ 17. Sales commissions.

_____ 18. Electricity costs of \$250 plus \$0.005 per kilowatt-hour after 1,000.

_____ 19. Quality control officer's salary.

_____ 20. Fabric used for the couch.

_____ 21. Oil used to lubricate sewing machines.

_____ 22. Straight-line depreciation on factory equipment.

_____ 23. Advertising cost of \$5,000 per month

_____ 24. Hourly wages of framers.

_____ 25. Foam used for stuffing the couch.

DuKray Concrete Company reported the following data for the month of March 2020. Beginning and ending inventories were negligible so were not considered. Prepare a variable costing Income Statement in the format provided.

	Total Cost	Number of Units	Unit Cost
Manufacturing Costs:			
Variable	\$ 288,000	20,000	\$ 54.40
Fixed	<u>512,000</u>	<u>20,000</u>	<u>25.60</u>
Total	<u>\$ 800,000</u>		<u>\$ 80.00</u>
Selling & Admin Expenses:			
Variable	\$3.50 per unit sold		
Fixed	\$ 125,000		
In the month of March, 18,000 units were sold at a price of \$120 per unit			

[illegible]

Problem 2 – Budgets (35 points)

Control Production requests your assistance in preparing budgets for the next year. Use the information below and formats on the next page to prepare a Sales Budget, Production Budget, and Direct Materials Budget.

- a. Estimated sales of the Small and Large Controls for the North, East, West, and South Divisions are as follows:

North Division

Small Controls	1,500 units at \$40 per unit
Large Controls	4,500 units at \$65 per unit

East Division

Small Controls	2,000 units at \$50 per unit
Large Controls	6,000 units at \$75 per unit

West Division

Small Controls	6,000 units at \$60 per unit
Large Controls	3,000 units at \$80 per unit

South Division

Small Controls	1,000 units at \$55 per unit
Large Controls	2,200 units at \$70 per unit

- b. Estimated beginning inventories and desired ending inventories:

	Beginning Inventory	Ending Inventory	Used in Production	Direct Mat Purchase Price
Direct Materials:				
Casing	280 units	300 units		
Wire	5,410 feet	4,500 feet		
Small Control				
Casing			1 per unit	\$9.10 per unit
Wire			20 feet	\$0.10 per foot
Large Control				
Casing			1 per unit	\$9.10 per unit
Wire			40 feet	\$0.10 per foot
Finished Goods:				
Small Controls	280	310		
Large Controls	330	350		

Control Productions Sales Budget For the year ending December 31, 2021	
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[illegible]

[illegible]

Problem 3 – Break Even Analysis (15 points)

Complete the following Break-Even situations. **Each option should start with the original assumptions**
(Round each answer to the nearest whole unit):

Unit Selling Price	\$1,200
Unit Variable Cost	800
Total Fixed Costs	415,000

1. Calculate the Break-Even point in units.
2. Assume the company is considering paying \$2,000 more per year to each of its **four** factory supervisor's salaries. What is the new Break-Even point in units?
3. What is the new Break-Even point be if cost of direct materials increased by \$50 per unit?
4. What would be the new Break-Even point is the selling price increased to \$1,280?
5. What is the sales volume in units needed to earn a target profit of \$500,000?